IU SCHOOL OF MEDICINE · OFFICE OF DATA SCIENCE AND AI

AI Education Initiatives

IUSM – Office of Data Science and AI

Marcus R. Breese, PhD

CCBB Retreat 2026

THE LANDSCAPE

AI training is spread across many units

UITS Research Technologies

HPC, GPU and LLM workshops

IU Research

AI Research Network; AI Research Brown Bag Talk

Indiana CTSI

Monthly AI Workshop Series for investigators

UITS Academic and AI Innovation

GenAI 101, office hours, faculty fellows

Schools and centers

Luddy AI Center, Kelley, IU Libraries

IU School of Medicine

Departments, student groups, ODSAI

ODSAI'S ROLE

IUSM's office to help you accelerate your research using AI

Not covered: IUSM's efforts on AI in clinical education.

TRAINING SERIES

Where IUSM researchers can start

Supercomputing for Everyone Series · UITS

LLMs on HPC; GPU Computing and AI on HPC

Series page: ittraining.iu.edu/explore-topics/titles/scfes

AI Workshop Series · Indiana CTSI

Next: Oct 27, competency-based AI training; Dec 8, AI and human subjects research (IRB)

Series page: indianactsi.org/research/data-informatics/ai-resources/ai-workshop-series

AI Research Brown Bag Talk · IU AI Research Network

Noon talks; next: Oct 26, Nov 9, Nov 23, Dec 7

All IU AI events: research.iu.edu/research-networks/ai

TOOLS YOU CAN USE

AI resources available at IU

Resource	Access	Notes
ChatGPT Edu	IU license	General chat and coding with Codex; restricted data OK; monthly credits go fast
Google Gemini / Gemini Notebook	IU access	General chat and light coding; restricted data OK
Microsoft Copilot	Azure	Restricted data OK; some costs may apply
Jetstream2 and NRP.ai	NSF resources	Open research that fits their allocation requirements; API access; no PHI or critical data on Jetstream2
REALLMS	UITS Research Technologies	The only option approved for PHI

TOOLS YOU CAN USE

REALLMS: IU's local LLMs

Models

- gpt-oss-120b
- Qwen3.8-27B
- glm5.3-flash (coming soon)

Access

- Chat interface
- API (through RT projects)
- Run on-premise by UITS Research Technologies

The only IU option approved for PHI

About REALLMS: researchdata.iu.edu/resource-catalog/about-reallms-at-iu

HELP WITH AI IN RESEARCH

Who to ask about AI in your research

ODSAI

- Focus: using AI to accelerate basic and computational research
- Education: how to actively apply AI tools to your research today
- Hands-on help when you need it
- Connects you with the right people

AI Governance Committee

- Members from IUSM, IU Health and Regenstrief
- Education, clinical, research and IT leaders
- Advisory group on running an AI research program the right way: approved tools, protected data, compliance

Dean’s office AI group

- Includes David Hanson, director of artificial intelligence
- Supports AI in clinical and education settings
- Proof-of-concept development and consulting

EXAMPLE · IU SCHOOL OF MEDICINE–NORTHWEST-GARY

A joint consult

The question

A group wanted to use AI tools to capture conversations in a clinical learning environment

The consult

ODSAI and the Dean’s office AI group met with the faculty members to discuss feasibility and implementation options for their project

IN THE CLASSROOM

Coordinating classroom efforts

Dayu Sun (Biostatistics) is developing a course on AI Literacy.

ODSAI's role

- Coordinate across classes with similar AI needs, so each class doesn't reinvent the wheel
- Share notes on AI tools and how to get students access
- For this course: computing needs, available LLMs for coursework

ODSAI'S OWN WORK

What makes biomedical AI different

Protected data

Patient data and IRB terms limit which tools can see what

Checking output

Fluent answers, citations and code can still be wrong

Reproducibility

Analyses still have to be rerun, reviewed and documented

Your own data

General tools don't know your lab's data, methods or literature

WE'RE PLANNING A HANDS-ON WORKSHOP

AI Tools for Biomedical Researchers

3

weekly sessions, 3.5 hours each

20–25

participants per session, with TAs

Hands-on

a personal VM using IU REALLMS local LLMs

Open to all IUSM research trainees (graduate students, postdocs, and clinical residents and fellows doing research rotations) and faculty.

THE WORKSHOP

Workshop goals: removing the magic

DAY 1

How AI works (and where it breaks)

Build a classifier, watch an LLM sample, trace an agent loop

DAY 2

Getting your data accessible to LLMs

A retrieval index and an MCP server connected to your own chat interface

DAY 3

A coding analysis exercise

Direct an AI coding harness (such as OpenCode or Codex) through a full RNA-seq analysis from a written plan with pre-registered checks

Three weekly 3.5-hour sessions · IUSM research trainees and faculty

WORKSHOP WEBSITE

ai-workshop.compgenlab.org

Thinking of using AI or LLM tools in your research? ODSAI can help.

We'll connect you with the right people.